Behnjamin Barlow

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OBJECTIVE

In a Scholarship For Service (SFS) degree path designated for Cyber Defense Education by the National Centers of Academic Excellence in cybersecurity. I have a deep interest in my cyber competitions like CCDC and platforms like Hack The Box and Pwn.College and hope to get a career in the cyber security field

EDUCATION

Tennessee Technological University (NCAE) Cookeville, TN (August 2022 - Present)

- Computer Science Bachelor of Engineering|Cybersecurity
- Computer Science Masters of Engineering|Cybersecurity (Exp. Fall 2027)

CERTIFICATIONS

Completed: CompTIA **Security+** **Expires:** 01/29

EXPERIENCE

CAISER Intern | Oak Ridge National Laboratory | (05/26/2026 - 07/31/2026) | Oak Ridge, TN

- Designed and implemented custom firmware and communication protocols between CubeSat and ground station hardware, enabling reliable telemetry capture for dataset generation
- Collected and structured sensor, battery, and RF telemetry data into training-ready datasets for evaluating small LM performance on satellite operations tasks
- Debugged and validated firmware behavior across sensor and power subsystems to ensure consistent data fidelity
- Collaborated with CAISER researchers to define target capabilities for small LMs, informing what telemetry/data needed to be captured from the testbed
- Documented testbed architecture and firmware design to support reproducibility and future dataset expansion

Cyber Range Application Developer | Tennessee Tech | (05/19/2025 - 08/01/2025) | Cookeville, TN

- Contributed to the development of Infrastructure as Code (IaC) solutions using Python, OpenTofu, and HashiCorp CDK
- Helped to automate the deployment and management of the CEROC Cyber Range on the Canonical MicroCloud hypervisor
- Worked closely with the Cyber Range Engineer to troubleshoot infrastructure problems and enhance overall system functionality

Satellite Security Research | Tennessee Tech | (12/1/2024 - Present) | Cookeville, TN

- Developing encrypted verification methods to ensure trusted communication built off the block chain model
- Designing and building a physical testbed simulating a satellite constellation using Jetson Orin devices as satellites and Raspberry Pis as ground stations to model communication and network interactions

Penetration Tester | ZOE International | (08/20/2025 - 01/16/2026) | Cookeville, TN

- Conducted a comprehensive security assessment of ZOE International public assets and delivered professional technical assessment reports
- Collaborated with internal stakeholders to explain technical risks in a business context, ensuring a path to remediation for the leadership team
- Utilized engagement frameworks such as Burp Suite, Metasploit, and more widely used industry tools.

COMPETITION EXPERIENCE

Application Security Lead | Collegiate Cyber Defense Competition Nationals (11/20/2024 - 4/25/2025)

- Secured web applications and databases by hardening configurations, managing access controls, and implementing network security policies
- Hunted, detected, and neutralized real-time Red Team attacks by analyzing network traffic, investigating logs, and mitigating active threats
- Administered Linux systems and pfSense firewalls, performing OS hardening, system auditing, and automating security tasks with Bash scripts
- Architected and deployed a 16+ machine training environment, simulating attack scenarios with vulnerable services (web, mail, domain)
- Executed offensive Red Team operations within the mock environment using SliverC2 and Realm to test persistence methods and improve defensive playbooks

Web App. Penetration Tester | Collegiate Penetration Testing Competition (08/16/2025 - 01/20/2026)

- Conducted comprehensive web application penetration tests on multi-tiered live environments, aligning findings with the OWASP Top 10 to identify and exploit critical flaws (e.g., SQL Injection, XSS, and authentication bypass)
- Developed a holistic attack narrative by collaborating with team members to correlate web application vulnerabilities with network-level findings, demonstrating full chain compromises from Layer 7 to Layer 3
- Authored professional-grade penetration test reports using Ghostwriter, detailing findings, analyzing business impact, and providing actionable, prioritized remediation strategies for development teams

Linux Subject Matter Expert | Tennessee Tech CCDC Nationals Team (12/01/2025 - 04/10/2026)

- Designed and deployed custom automation scripts in go and bash to rapidly enforce security policy, audit user accounts, and implement kernel-level hardening across multiple Linux machines.
- Established proactive threat hunting and forensic capabilities to detect advanced Red Team persistence, focusing on tracking covert beacons, kernel hooks, and user-space rootkits.
- Engineered dynamic, isolated testing environments featuring Red Team tactics and compromised machines to validate hardening scripts and incident response procedures in a live-fire competition scenario.
- Secured critical Linux services by implementing secure configurations, strong host-based firewall policies, and file integrity monitoring.

SKILLS

- Offensive Cyber Tools: Mythic C2, SliverC2, Kali Linux, Burp Suite, Hashcat/John, Nmap, Persistence Techniques, Metasploit.
- Defensive & Forensics/Rev Tools: WireShark, Autopsy, FTK Imager, Iptables, PfSense, SIEM, Advanced File Integrity Monitoring, Kernel Hooks, User-Space Rootkits, GDB, PwnDBG, Ghidra, Binary ninja.
- Programming Languages: Python, Bash, C++, HTML, CSS, JavaScript, GoLang, PowerShell.
- Databases: MySQL, MongoDB. Sqlite3, PostgreSQL.
- Virtualization & Cloud: Proxmox VE, VMware, ESXi, Google Cloud Platform, Canonical, OpenTofu, HashiCorp CDK.
- Infrastructure & Automation: Ansible, Salt Stack, Configuration Management, Git, GitHub, GitLab, GitKraken, TailScale.
- Operating Systems: Linux (Hardening/Administration), Windows.
- Advanced Networking: Virtual Networking, DNS, VPN, Network Troubleshooting, CAN Bus/Internal Protocols.
- Specialized Concepts: Data Forensics, Computer Forensics, NTA (Network Traffic Analysis), Web Security.

LEADERSHIP EXPERIENCE

Agile Scrum Master | Tennessee Tech School of Music Adjudication System (04/01/2025 - 12/09/2025)

- Served as Scrum Master for a 6-person team , coordinating all sprints, managing the product backlog, and ensuring the timely delivery of a fully digital music jury scheduling and evaluation platform
- Oversaw the design and implementation of security features, including Azure Active Directory (Azure AD) authentication for faculty , Role-Based Access Control (RBAC) via SharePoint Groups, and strict Least-Privilege Enforcement , ensuring compliance with institutional data management standards.
- Successfully developed the platform entirely within the university's existing Microsoft 365 licensing (SharePoint Online, Power Automate, Forms, Bookings, Teams) , delivering a fully digital, integrated, and sustainable solution that eliminated the need for expensive third-party platforms and reduced administrative workload.
- Directed the design of a NoSQL/List-based data model and managed the core Power Automate workflows and Excel Office Scripts responsible for automated data routing and feedback generation

Tennessee Tech Cyber Eagles Volunteer | President (05/01/2025 - Present)

- Engineering and Deployed the Official Club Website (cybereagles.org), establishing a permanent static site using free hosting to ensure technical longevity and resource efficiency.
- Coordinated and Hosted regular club meetings, securing engagement from guest speakers from leading technology companies and government agencies.
- Developed and executed outreach initiatives with related academic departments and student groups, successfully increasing club visibility and membership.
- Managed administrative operations, partnering with university administration and student organizations to manage budgets, secure necessary funding, and ensure smooth club operations.

Tennessee Tech Cyber Interest Group Mentor (09/15/2024 - 05/08/2026)

- Developed and Led Technical Training Sessions on complex cybersecurity topics, including digital forensics, privilege escalation, and rootkits, preparing students for real-world offensive and defensive security challenges
- Designed and Delivered Educational Content, including detailed meeting agendas, presentations (slides), and practical labs to clarify concepts for a diverse audience
- Cultivated a Dynamic, Collaborative Learning Environment for students across all skill levels, significantly improving group knowledge and practical application skills in areas like network defense and ethical hacking
- Utilized LXD cyber range with SaltStack to create secure, repeatable, and realistic training environments for hands-on practice, mirroring professional cybersecurity lab setups
- Collaborated with fellow mentors and group leads to strategize meeting schedules and content flow, ensuring timely and smooth execution of all training sessions

PROJECTS

Personal Website:

- Established and Maintained a permanent static website, utilizing free software hosting solutions to ensure optimal resource efficiency.
- Configured a custom domain to serve the site, centralizing professional content, public code repositories (GitHub), and academic projects.
- The site serves as a portfolio Including insights into cybersecurity work, showcasing technical proficiency and professional contribution

Cyber Truck Challenge (06/01/2025 - 06/09/2025) | Battle Creek, MI

- Performed specialized security assessments targeting vehicle communication systems and embedded hardware, including analysis of CAN Bus and internal protocols.
- Utilized deep analysis techniques to exploit security vulnerabilities in the simulated automotive environment.

Home Lab:

- Managing a high-performance virtualization cluster using a Proxmox hypervisors, to support resource-intensive cybersecurity operations.
- Established secure, isolated virtual ranges dedicated to specialized training and testing, including local malware analysis, Command and Control (C2) testing, and execution of complex Linux scripting tasks
- Supported competitive cybersecurity training (CCDC) by designing and deploying complex server infrastructure and mock competition environments for team exercises
- Designed and deployed a dedicated Raspberry Pi appliance to function as a Tailscale router and VPN endpoint, enabling secure, encrypted remote access to the entire lab network
- Automated lab functionality by implementing Wake-on-LAN (WoL) via the Raspberry Pi, allowing remote power cycling of the Proxmox servers to optimize energy consumption and accessibility